

Final Ref Comments

April 12, 2026

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Dependent Variable Specifications

Table 1: Main Table

	C. Goodman		Census of Governments		Census
	Municipalities		School districts	Special Districts	Main City Share
	(1)	(2)	(3)	(4)	(5)
Panel A: First Stage					
$\widehat{GM}$	1.771*** (0.315)	1.771*** (0.315)	1.841*** (0.330)	1.771*** (0.315)	1.771*** (0.315)
Panel B: OLS 1940-1970					
GM	0.003 (0.003)	0.005 (0.004)	0.101 (0.077)	-0.006 (0.009)	-0.896*** (0.171)
Panel C: 2SLS 1940-1970					
GM	0.008*** (0.003)	0.011*** (0.003)	0.177** (0.072)	0.011* (0.006)	-1.577*** (0.090)
1940-70 Avg.	-0.26	-0.33	-12.95	0.64	-3.37
Panel D: OLS 1940-2010					
GM	0.005 (0.005)	0.007 (0.005)	0.101 (0.077)	-0.009 (0.008)	-1.137*** (0.223)
Panel E: 2SLS 1940-2010					
GM	0.010*** (0.003)	0.012*** (0.004)	0.178** (0.073)	0.020*** (0.006)	-1.873*** (0.130)
1940-2010 Avg.	-0.39	-0.49	-13.31	1.04	-7.96
1940 Avg.	1.49	1.61	14.09	0.89	32.86
First Stage F-Stat	31.55	31.55	31.18	31.55	31.55
Observations	130	130	118	130	130

Table 2: Ref Request 1: Log Differences

	C. Goodman		Census of Governments		Census
	Municipalities		School districts	Special Districts	Main City Share
	(1)	(2)	(3)	(4)	(5)
Panel A: First Stage					
$\hat{GM}$	1.771*** (0.315)	1.771*** (0.315)	1.842*** (0.330)	1.761*** (0.326)	1.771*** (0.315)
Panel B: OLS 1940-1970					
GM	0.003 (0.003)	0.005 (0.003)	0.005 (0.017)	0.004 (0.020)	-0.017*** (0.006)
Panel C: 2SLS 1940-1970					
GM	0.002 (0.002)	0.005 (0.003)	-0.031*** (0.009)	0.003 (0.008)	-0.039*** (0.004)
1940-70 Avg.	0.13	0.08	-1.80	1.15	0.27
Panel D: OLS 1940-2010					
GM	0.003 (0.003)	0.000 (0.005)	0.003 (0.017)	-0.017 (0.018)	-0.035*** (0.010)
Panel E: 2SLS 1940-2010					
GM	0.000 (0.003)	0.000 (0.004)	-0.036*** (0.010)	-0.023** (0.011)	-0.068*** (0.006)
1940-2010 Avg.	0.18	0.12	-1.92	1.71	0.30
1940 Avg.	1.48	1.61	14.08	0.88	3282.25
First Stage F-Stat	31.54	31.54	31.18	29.18	31.54
Observations	130	130	118	114	130

Table 3: Ref Request 1: Log Differences + popgrowth control

	C. Goodman		Census of Governments		Census
	Municipalities		School districts	Special Districts	Main City Share
	(1)	(2)	(3)	(4)	(5)
Panel A: First Stage					
$\hat{GM}$	1.865*** (0.319)	1.865*** (0.319)	1.939*** (0.339)	1.852*** (0.331)	1.865*** (0.319)
Panel B: OLS 1940-1970					
GM	0.003 (0.003)	0.005 (0.003)	0.006 (0.017)	0.004 (0.021)	-0.017*** (0.006)
Panel C: 2SLS 1940-1970					
GM	0.004** (0.002)	0.007** (0.003)	-0.023** (0.009)	0.003 (0.007)	-0.030*** (0.002)
1940-70 Avg.	0.13	0.08	-1.80	1.15	0.27
Panel D: OLS 1940-2010					
GM	0.003 (0.003)	0.000 (0.005)	0.003 (0.017)	-0.018 (0.018)	-0.035*** (0.009)
Panel E: 2SLS 1940-2010					
GM	0.004 (0.003)	0.004 (0.004)	-0.028*** (0.010)	-0.017* (0.010)	-0.054*** (0.004)
1940-2010 Avg.	0.18	0.12	-1.92	1.71	0.30
1940 Avg.	1.48	1.61	14.08	0.88	3282.25
First Stage F-Stat	34.18	34.18	32.73	31.39	34.18
Observations	130	130	118	114	130

Table 4: Ref Request 2: Pct Outcome

	C. Goodman		Census of Governments		Census
	Municipalities		School districts	Special Districts	Main City Share
	(1)	(2)	(3)	(4)	(5)
Panel A: First Stage					
$\hat{GM}$	1.771*** (0.315)	1.771*** (0.315)	1.842*** (0.330)	1.771*** (0.315)	1.771*** (0.315)
Panel B: OLS 1940-1970					
GM	-0.000 (0.002)	0.002 (0.003)	0.097 (0.076)	-0.000 (0.015)	-115.435*** (39.031)
Panel C: 2SLS 1940-1970					
GM	0.002 (0.002)	0.006** (0.002)	0.159** (0.068)	0.051*** (0.014)	-313.303*** (35.240)
1940-70 Avg.	0.16	0.11	-12.40	1.40	1722.81
Panel D: OLS 1940-2010					
GM	-0.001 (0.003)	0.000 (0.004)	0.094 (0.077)	0.002 (0.023)	-226.522*** (77.510)
Panel E: 2SLS 1940-2010					
GM	0.002 (0.003)	0.004 (0.003)	0.151** (0.069)	0.112*** (0.029)	-663.575*** (99.163)
1940-2010 Avg.	0.24	0.15	-12.66	2.87	3449.08
1940 Avg.	1.48	1.61	14.08	0.88	3282.25
First Stage F-Stat	31.54	31.54	31.18	31.54	31.54
Observations	130	130	118	130	130

Table 5: Ref Request 2: Pct on Pct

	C. Goodman	Census of Governments			Census
	Municipalities		School districts	Special Districts	Main City Share
	(1)	(2)	(3)	(4)	(5)
Panel A: First Stage					
$\hat{GM}$	0.432 (0.294)	0.432 (0.294)	0.367 (0.346)	0.432 (0.294)	0.432 (0.294)
Panel B: OLS 1940-1970					
Percentage Change in Urban Black Population	-0.004** (0.002)	-0.002 (0.002)	0.038 (0.091)	0.007 (0.018)	58.479 (54.331)
Panel C: 2SLS 1940-1970					
Percentage Change in Urban Black Population	0.008 (0.007)	0.026** (0.011)	0.797 (0.565)	0.211** (0.098)	-1.3e+03*** (482.842)
1940-70 Avg.	0.16	0.11	-12.40	1.40	1722.81
Panel D: OLS 1940-2010					
Percentage Change in Urban Black Population	-0.006** (0.003)	-0.005* (0.003)	0.034 (0.092)	-0.002 (0.028)	107.667 (129.110)
Panel E: 2SLS 1940-2010					
Percentage Change in Urban Black Population	0.006 (0.010)	0.016 (0.013)	0.759 (0.552)	0.459** (0.198)	-2.7e+03** (1074.699)
1940-2010 Avg.	0.24	0.15	-12.66	2.87	3449.08
1940 Avg.	1.48	1.61	14.08	0.88	3282.25
First Stage F-Stat	2.15	2.15	1.13	2.15	2.15
Observations	130	130	118	130	130

Table 6: Raw Differences

	C. Goodman		Census of Governments		Census
	Municipalities		School districts	Special Districts	Main City Share
	(1)	(2)	(3)	(4)	(5)
Panel A: First Stage					
$\hat{GM}$	1.771*** (0.315)	1.771*** (0.315)	1.842*** (0.330)	1.771*** (0.315)	1.771*** (0.315)
Panel B: OLS 1940-1970					
GM	1.148** (0.575)	1.103* (0.619)	-16.922** (7.465)	8.039* (4.531)	-1.5e+04*** (5152.559)
Panel C: 2SLS 1940-1970					
GM	1.740*** (0.392)	1.824*** (0.436)	-26.682*** (4.893)	9.481*** (1.628)	-3.4e+04*** (5930.385)
1940-70 Avg.	7.22	5.39	-320.96	52.42	48923.45
Panel D: OLS 1940-2010					
GM	1.353* (0.774)	1.027 (0.810)	-17.575** (7.579)	6.666 (4.188)	-4.5e+04*** (8692.185)
Panel E: 2SLS 1940-2010					
GM	1.821*** (0.590)	1.839*** (0.630)	-27.567*** (4.987)	14.356*** (2.209)	-7.8e+04*** (1.1e+04)
1940-2010 Avg.	10.34	7.67	-327.23	93.10	64430.39
1940 Avg.	1.48	1.61	14.08	0.88	3282.25
First Stage F-Stat	31.54	31.54	31.18	31.54	31.54
Observations	130	130	118	130	130

Table 7: 1970 P.C. Outcome

	C. Goodman		Census of Governments		Census
	Municipalities		School districts	Special Districts	Main City Share
	(1)	(2)	(3)	(4)	(5)
Panel A: First Stage					
$\hat{GM}$	1.762*** (0.310)	1.762*** (0.308)	1.824*** (0.336)	1.788*** (0.321)	1.648*** (0.328)
Panel B: OLS 1940-1970					
GM	0.002 (0.002)	0.003 (0.003)	-0.001 (0.004)	-0.005 (0.009)	-71.272*** (17.313)
Panel C: 2SLS 1940-1970					
GM	0.012*** (0.004)	0.015*** (0.004)	0.005 (0.004)	0.053*** (0.014)	-136.271*** (9.157)
1940-70 Avg.	1.22	1.28	1.13	1.52	2946.69
Panel D: OLS 1940-2010					
GM	0.004 (0.004)	0.004 (0.004)	-0.003 (0.003)	-0.007 (0.009)	-81.939*** (15.204)
Panel E: 2SLS 1940-2010					
GM	0.014*** (0.004)	0.015*** (0.005)	0.005** (0.002)	0.066*** (0.012)	-143.265*** (9.955)
1940-2010 Avg.	1.09	1.13	0.78	1.93	2489.56
1940 Avg.	1.48	1.61	14.08	0.88	3282.25
First Stage F-Stat	32.26	32.74	29.42	30.99	25.27
Observations	130	130	118	130	130

Table 8: 1970 Outcome

	C. Goodman		Census of Governments		Census
	Municipalities		School districts	Special Districts	Main City Share
	(1)	(2)	(3)	(4)	(5)
Panel A: First Stage					
$\hat{GM}$	1.398*** (0.308)	1.377*** (0.300)	1.587*** (0.284)	1.623*** (0.305)	1.758*** (0.304)
Panel B: OLS 1940-1970					
GM	-0.221 (0.563)	-0.321 (0.621)	6.245** (2.591)	5.932 (3.986)	-1.7e+04*** (5186.785)
Panel C: 2SLS 1940-1970					
GM	5.552*** (0.819)	5.978*** (0.847)	3.536** (1.558)	10.958*** (2.027)	-3.9e+04*** (5443.906)
1940-70 Avg.	56.09	58.35	57.94	83.74	310000.00
Panel D: OLS 1940-2010					
GM	-0.490 (0.848)	-0.647 (0.882)	6.028** (2.426)	3.286 (2.794)	-4.7e+04*** (9808.605)
Panel E: 2SLS 1940-2010					
GM	5.512*** (0.970)	5.797*** (0.974)	2.886* (1.482)	15.223*** (2.588)	-8.3e+04*** (1.0e+04)
1940-2010 Avg.	59.21	60.63	51.67	124.42	330000.00
1940 Avg.	1.48	1.61	14.08	0.88	3282.25
First Stage F-Stat	20.57	20.99	31.24	28.40	33.48
Observations	130	130	118	130	130

Table 9: 1970 Outcome, Popgrowth Control

	C. Goodman		Census of Governments		Census
	Municipalities		School districts	Special Districts	Main City Share
	(1)	(2)	(3)	(4)	(5)
Panel A: First Stage					
$\hat{GM}$	1.493*** (0.313)	1.470*** (0.306)	1.667*** (0.295)	1.718*** (0.305)	1.851*** (0.308)
Panel B: OLS 1940-1970					
GM	-0.244 (0.570)	-0.342 (0.626)	6.320** (2.474)	5.996 (4.050)	-1.7e+04*** (4595.081)
Panel C: 2SLS 1940-1970					
GM	5.388*** (0.820)	5.830*** (0.847)	4.554*** (1.442)	10.001*** (1.967)	-3.3e+04*** (4682.853)
1940-70 Avg.	56.09	58.35	57.94	83.74	310000.00
Panel D: OLS 1940-2010					
GM	-0.531 (0.856)	-0.678 (0.897)	6.111*** (2.292)	3.251 (2.802)	-4.7e+04*** (9230.720)
Panel E: 2SLS 1940-2010					
GM	5.435*** (0.966)	5.701*** (0.971)	3.954*** (1.355)	14.940*** (2.555)	-7.1e+04*** (8702.952)
1940-2010 Avg.	59.21	60.63	51.67	124.42	330000.00
1940 Avg.	1.48	1.61	14.08	0.88	3282.25
First Stage F-Stat	22.69	23.13	31.82	31.62	36.03
Observations	130	130	118	130	130

Table 10: Poisson

	C. Goodman		Census of Governments	
	Municipalities		School districts	Special Districts
	(1)	(2)	(3)	(4)
Panel A: First Stage				
$\hat{GM}$	1.398*** (0.308)	1.377*** (0.300)	1.587*** (0.283)	1.623*** (0.305)
Panel B: OLS 1940-1970				
GM	0.016*** (0.006)	0.016** (0.006)	0.052*** (0.018)	0.039* (0.021)
Panel C: 2SLS 1940-1970				
GM	0.012 (0.011)	0.012 (0.011)	0.003 (0.025)	0.039 (0.034)
1940-70 Avg.	7.22	5.39	-320.96	52.42
Panel D: OLS 1940-2010				
GM	0.015** (0.007)	0.013* (0.007)	0.052*** (0.017)	0.016 (0.012)
Panel E: 2SLS 1940-2010				
GM	0.010 (0.013)	0.012 (0.013)	0.002 (0.024)	0.039 (0.025)
1940-2010 Avg.	10.34	7.67	-327.23	93.10
1940 Avg.	48.87	52.96	381.55	31.32
First Stage F-Stat	20.57	20.99	31.51	28.40
Observations	130	130	117	130

Table 11: Poisson, popgrowth control

	C. Goodman		Census of Governments	
	Municipalities		School districts	Special Districts
	(1)	(2)	(3)	(4)
Panel A: First Stage				
$\hat{GM}$	1.493*** (0.313)	1.470*** (0.306)	1.667*** (0.294)	1.718*** (0.305)
Panel B: OLS 1940-1970				
GM	0.016*** (0.006)	0.016*** (0.006)	0.054*** (0.016)	0.038* (0.021)
Panel C: 2SLS 1940-1970				
GM	0.016* (0.010)	0.017* (0.010)	0.011 (0.023)	0.032 (0.031)
1940-70 Avg.	7.22	5.39	-320.96	52.42
Panel D: OLS 1940-2010				
GM	0.014** (0.007)	0.013* (0.007)	0.055*** (0.015)	0.016 (0.012)
Panel E: 2SLS 1940-2010				
GM	0.016 (0.012)	0.017 (0.011)	0.011 (0.022)	0.042* (0.024)
1940-2010 Avg.	10.34	7.67	-327.23	93.10
1940 Avg.	48.87	52.96	381.55	31.32
First Stage F-Stat	22.69	23.13	32.10	31.62
Observations	130	130	117	130

Table 12: Poisson, 1970 Population exposure

	C. Goodman		Census of Governments	
	Municipalities		School districts	Special Districts
	(1)	(2)	(3)	(4)
Panel A: First Stage				
$\hat{GM}$	1.743*** (0.320)	1.740*** (0.323)	1.857*** (0.345)	1.800*** (0.325)
Panel B: OLS 1940-1970				
GM	0.003 (0.004)	0.005 (0.005)	0.009 (0.008)	0.018 (0.014)
Panel C: 2SLS 1940-1970				
GM	0.023*** (0.006)	0.027*** (0.007)	-0.002 (0.016)	0.053 (0.035)
1940-70 Avg.	7.22	5.39	-320.96	52.42
Panel D: OLS 1940-2010				
GM	0.003 (0.005)	0.004 (0.005)	0.006 (0.008)	-0.007 (0.009)
Panel E: 2SLS 1940-2010				
GM	0.021*** (0.006)	0.025*** (0.007)	-0.005 (0.016)	0.053* (0.030)
1940-2010 Avg.	10.34	7.67	-327.23	93.10
1940 Avg.	48.87	52.96	381.55	31.32
First Stage F-Stat	29.62	29.03	28.96	30.66
Observations	130	130	117	130

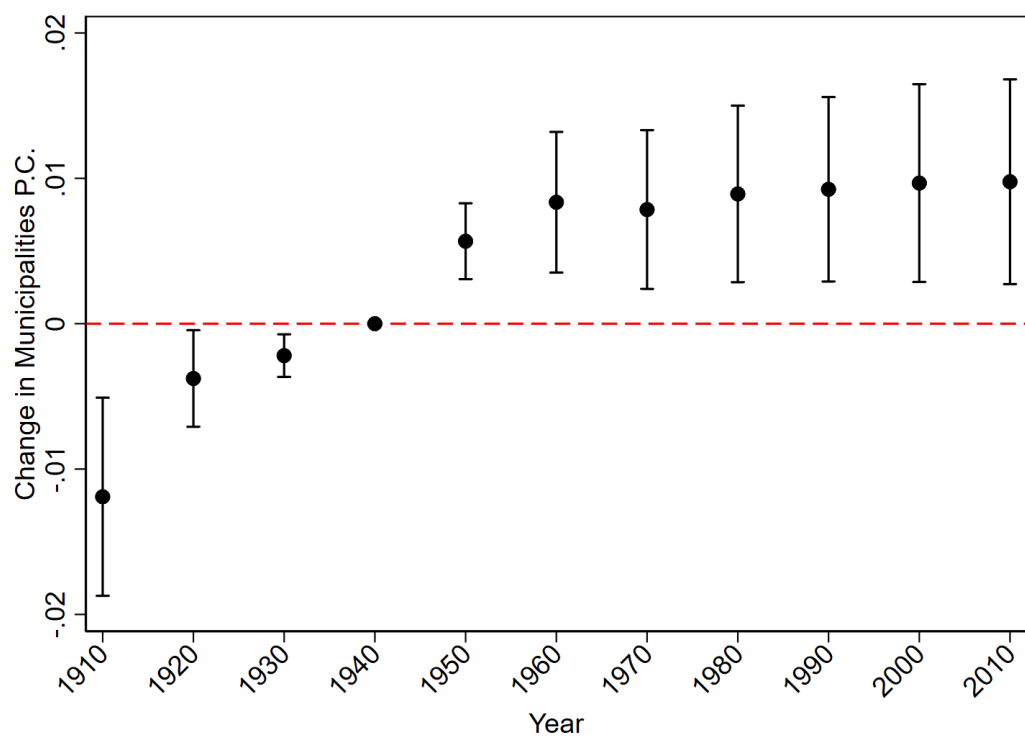


Figure 1: 1940 Imbalanced Controls

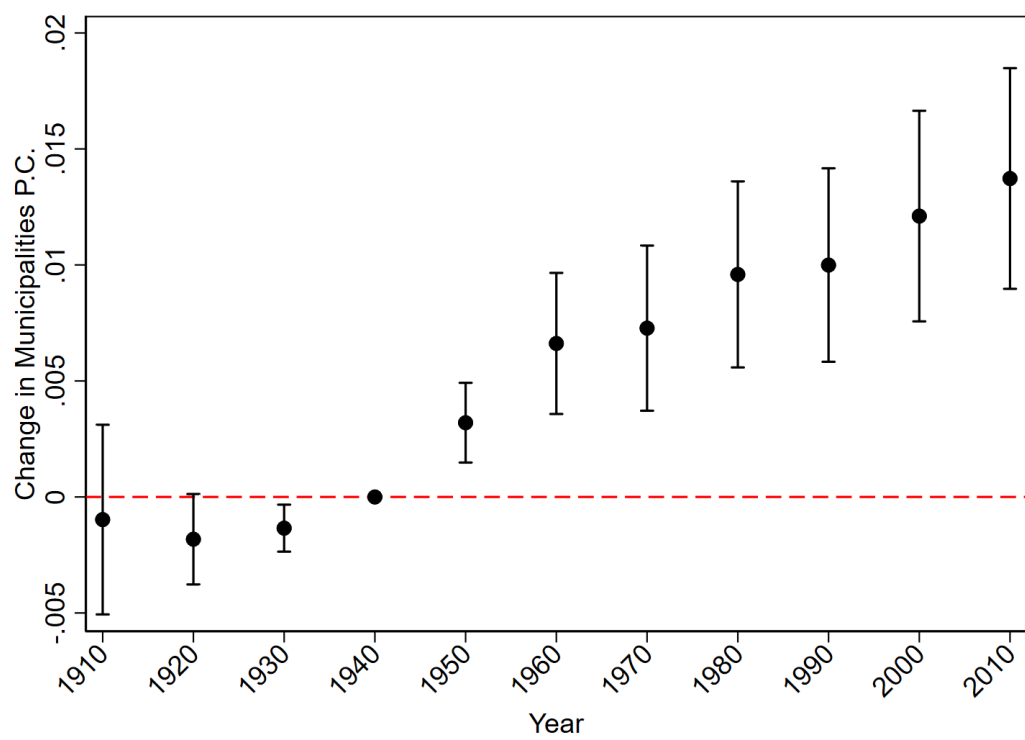


Figure 2: Baseline Controls

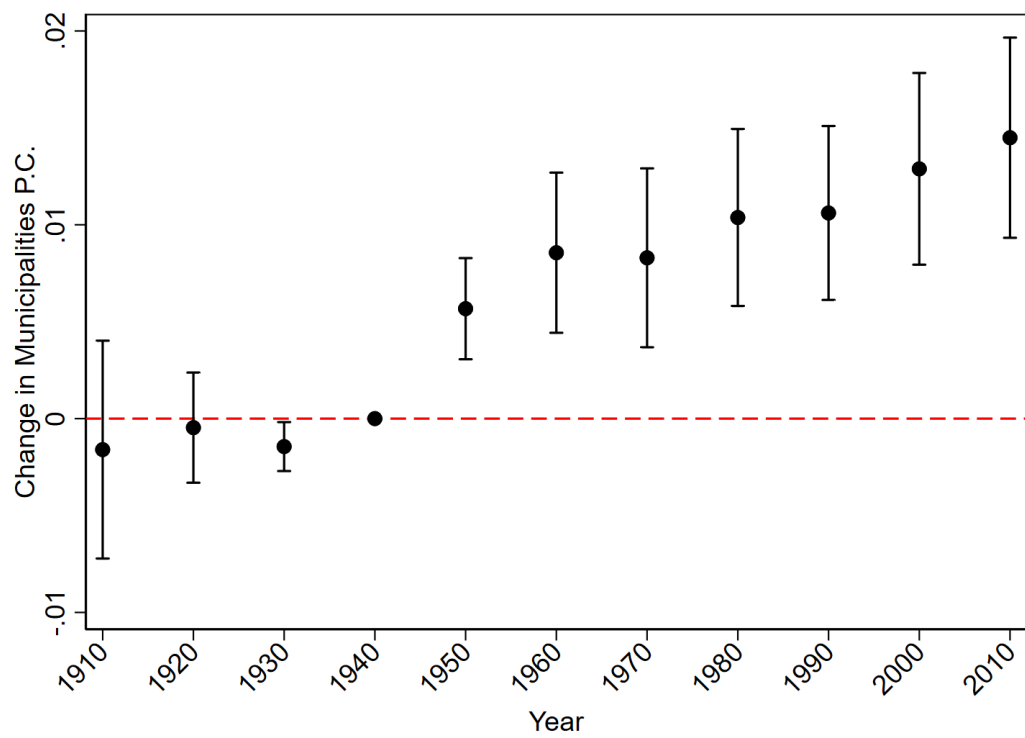


Figure 3: 1940 Mean Income, Lab. Force. Mfg and Popdens Varying

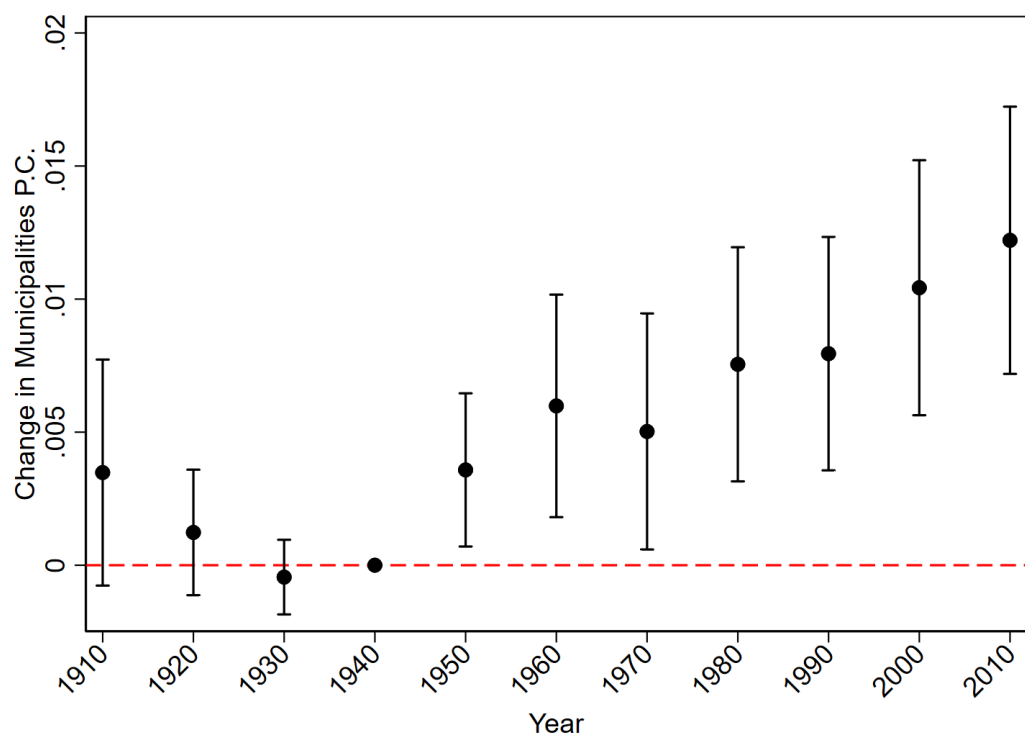


Figure 4: Lab. Force. Mfg and Popdens Varying